

STAT 566 Final Project

Andrew (Andy) Shin (ID: 1937870)

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```
rhc <- ATbounds::RHC

# rhc$survival <- factor(rhc2$survival)
# rhc$RHC <- factor(rhc$RHC)

# Install necessary packages if you haven't already
# install.packages(c("dagitty", "ggdag", "ggplot2"))
# Define the DAG and set coordinates to match the image layout
rhc_dag <- dagitty("dag {
  cat1 -> pafi1;
  cat1 -> scoma1;
  cat1 -> RHC;
  age -> chfhx;
  age -> dnr1_Yes;
  age -> meanbp1;
  chfhx -> survival;
  chfhx -> meanbp1;
  chfhx -> pafi1;
  meanbp1 -> aps1;
  meanbp1 -> survival;
  meanbp1 -> pafi1;
  meanbp1 -> scoma1;
  meanbp1 -> RHC;
  pafi1 -> aps1;
  pafi1 -> dnr1_Yes;
  pafi1 -> RHC;
  scoma1 -> aps1;
  scoma1 -> dnr1_Yes;
  aps1 -> survival;
  aps1 -> RHC;
  dnr1_Yes -> survival;
  RHC -> survival
}")

coordinates(rhc_dag) <- list(
  x = c(cat1 = 0, age = 0, chfhx = 0,
        pafi1 = 1, meanbp1 = 1, scoma1 = 1,
        aps1 = 2, dnr1_Yes = 2,
        RHC = 3, survival = 3),

  # Staggered Y-coordinates to dodge straight lines
  y = c(cat1 = 0, age = 2, chfhx = 4,
```

```

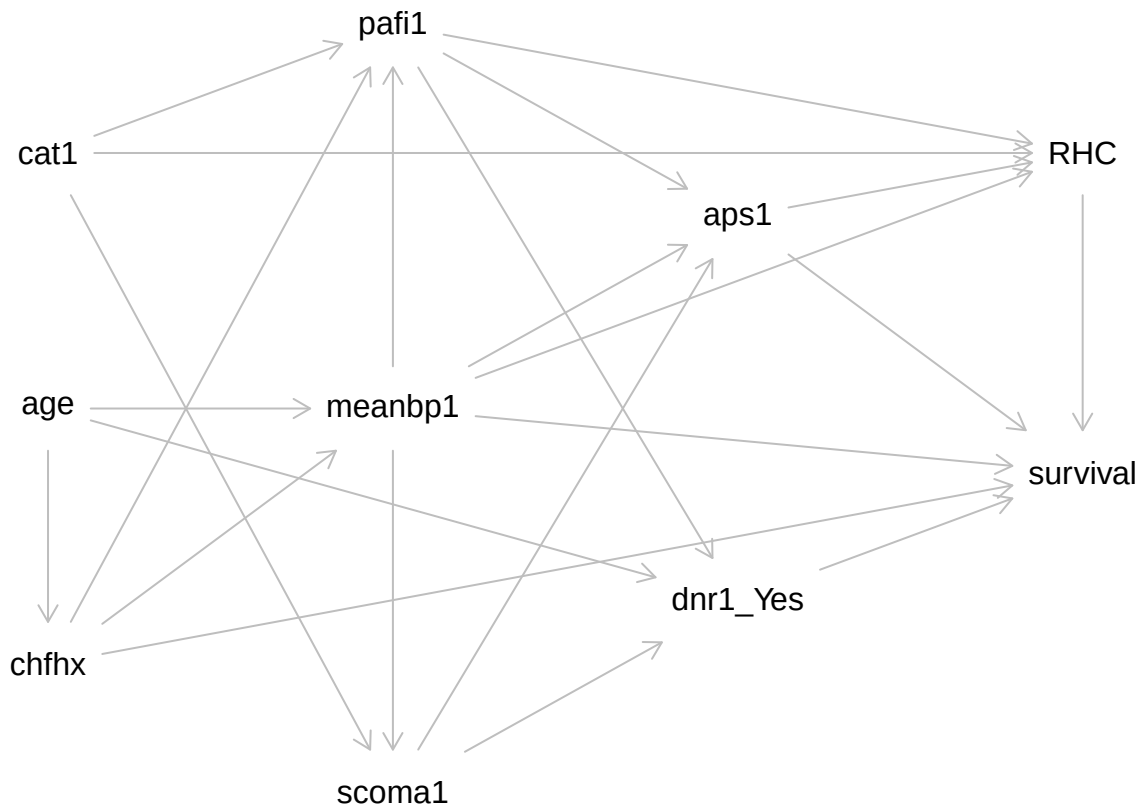
    pafi1 = -1, meanbp1 = 2, scoma1 = 5,
    aps1 = 0.5, dnr1_Yes = 3.5,
    RHC = 0, survival = 2.5)
)

```

```

# 1. Base dagitty plot
plot(rhc_dag)

```

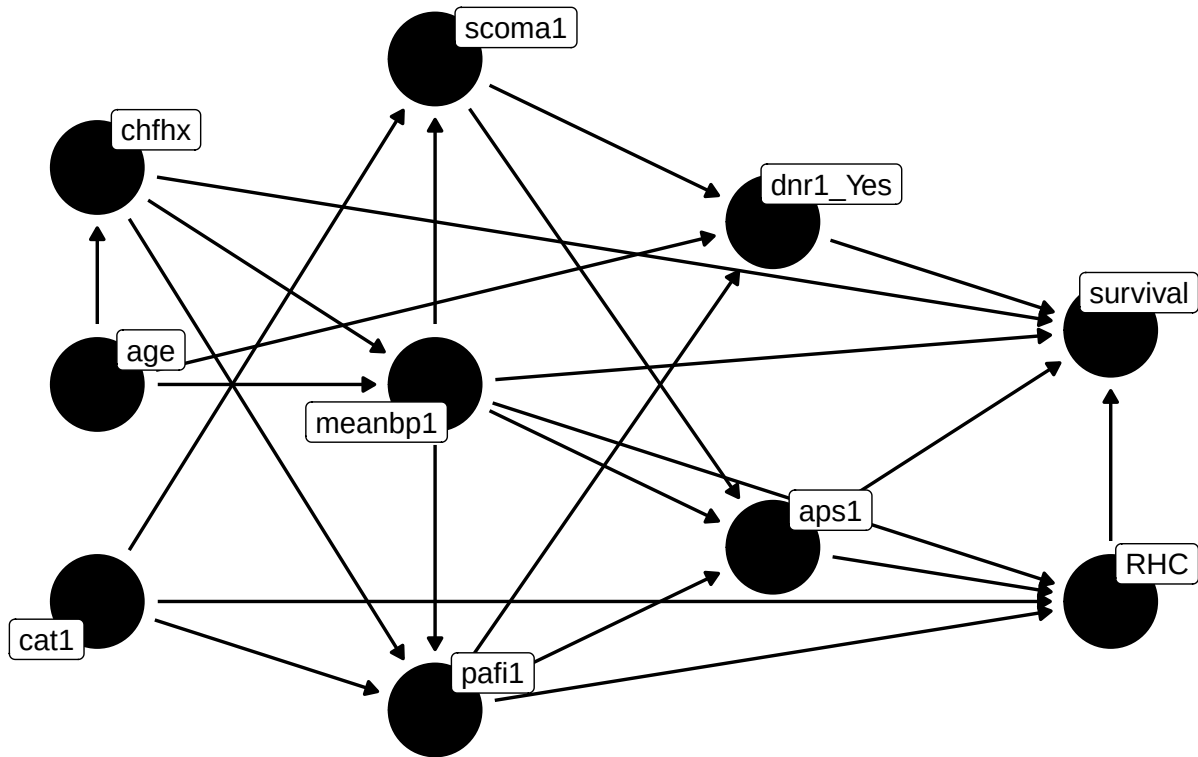


```

# 2. Presentation-ready plot using ggdag
ggdag(rhc_dag, text = FALSE, use_labels = "name") +
  theme_dag() +
  ggtitle("Figure 2: DAG model of RHC study") +
  theme(plot.title = element_text(hjust = 0.5))

```

Figure 2: DAG model of RHC study



RHC: Right Heart Catheterization (RHC) status during the first 24 hours (Treatment, A).

survival: All-cause mortality up to 180 days (Outcome, Y).

age: Patient age in years.

cat1: Primary disease category prompting ICU admission.

chfhx: Baseline history of Congestive Heart Failure.

meanbp1: Mean blood pressure recorded on day 1.

pafi1: Ratio of arterial oxygen partial pressure to fractional inspired oxygen ($\text{PaO}_2/\text{FIO}_2$) on day 1.

scoma1: Glasgow Coma Score on day 1.

aps1: APACHE III score on day 1.

dnr1_yes: Do Not Resuscitate (DNR) status established on day 1.

```
adjustmentSets(rhc_dag, exposure = "RHC", outcome = "survival")
```

```
## { aps1, chfhx, dnr1_Yes, meanbp1 }
## { aps1, chfhx, meanbp1, pafi1, scoma1 }
## { aps1, cat1, meanbp1, pafi1 }
```

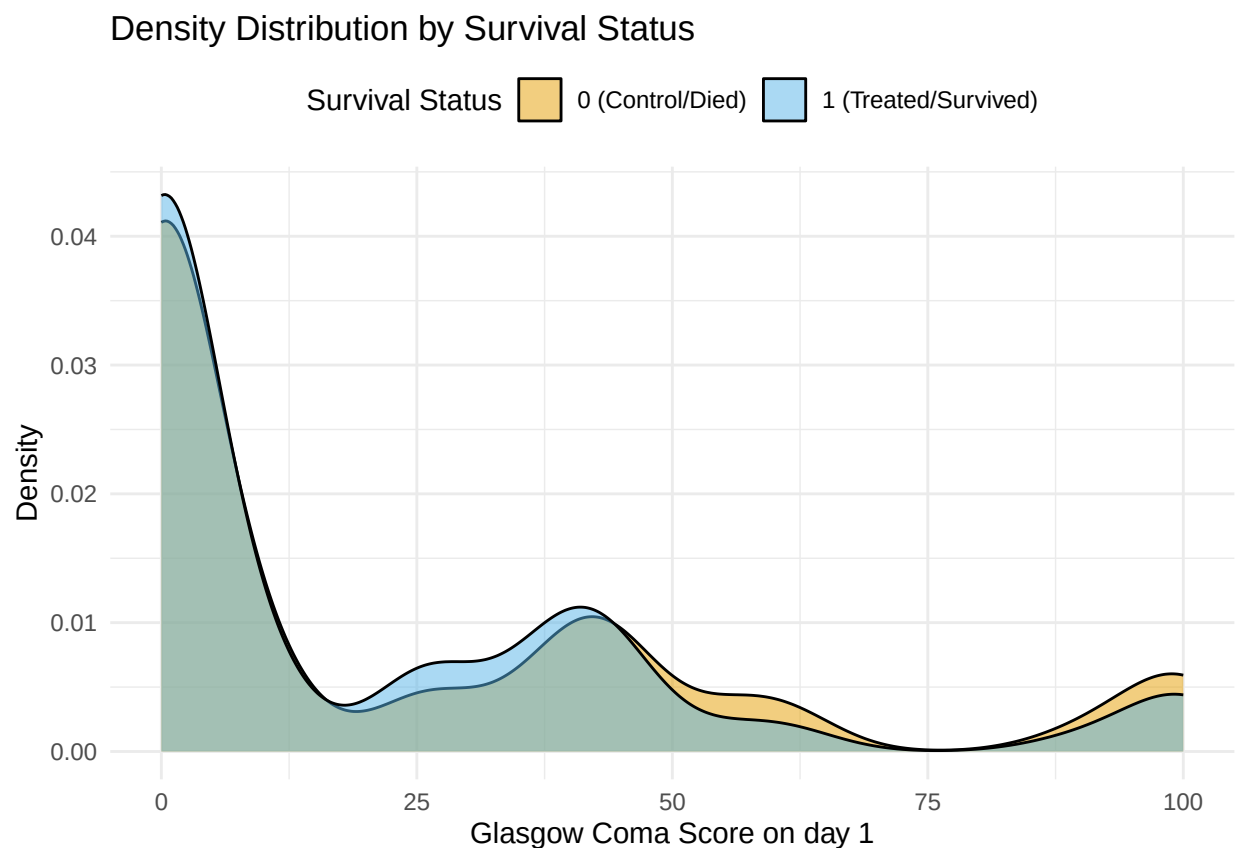
```
library(ggplot2)
```

```
# Ensure survival is treated as a categorical factor, not a continuous integer
rhc$survival <- factor(rhc$survival, levels = c(0, 1))
```

```
ggplot(rhc, aes(x = scoma1, fill = factor(RHC))) +
  # alpha = 0.5 makes the distributions semi-transparent so you can see the overlap
  geom_density(alpha = 0.5) +

  # Optional: Define clean, colorblind-friendly colors and update legend labels
  scale_fill_manual(
    values = c("0" = "#E69F00", "1" = "#56B4E9"),
    labels = c("0 (Control/Died)", "1 (Treated/Survived)")
  ) +

  # Clean, publication-ready theme
  theme_minimal() +
  labs(
    title = "Density Distribution by Survival Status",
    x = "Glasgow Coma Score on day 1",
    y = "Density",
    fill = "Survival Status"
  ) +
  theme(legend.position = "top")
```



```
rhc <- rhc %>%
  mutate(
    cat1 = case_when(
      cat1_CHF == 1 ~ "CHF",
      cat1_Cirrhosis == 1 ~ "Cirrhosis",
```

```

cat1_Colon_Cancer == 1 ~ "Colon Cancer",
cat1_Coma == 1 ~ "Coma",
cat1_COPD == 1 ~ "COPD",
cat1_Lung_Cancer == 1 ~ "Lung Cancer",
cat1_MOSF_Malignancy == 1 ~ "MOSF Malignancy",
cat1_MOSF_Sepsis == 1 ~ "MOSF Sepsis",
TRUE ~ "ARF" # If all above are 0, it defaults to ARF
)
)

```

```

cat_vars <- c("aps1", "chfhx", "dnr1_Yes", "meanbp1")
survival_form <- as.formula(paste("survival ~ ",
                                paste(cat_vars, collapse = "+")))

fit_mu0 <- glm(survival_form, data = subset(rhc, RHC == 0), family=binomial)
mu0_hat <- predict(fit_mu0, newdata = rhc, type="response")
tau_ATT_reg <- mean(rhc$survival[rhc$RHC==1] - mu0_hat[rhc$RHC == 1])

```

```

## Warning in Ops.factor(rhc$survival[rhc$RHC == 1], mu0_hat[rhc$RHC == 1]): '-
## not meaningful for factors

```

```

cat("ATT via outcome regression: ", round(tau_ATT_reg, 3), "\n")

```

```

## ATT via outcome regression: NA

```

```

e_hat <- predict(glm(RHC ~ chfhx + aps1 + meanbp1 + dnr1_Yes, data = rhc, family = binomial), type = "r
w_treat <- rhc$RHC + (1 - rhc$RHC) * (e_hat / (1 - e_hat))
tau_ATT_IPW <- mean(rhc$survival[rhc$RHC == 1]) - (sum((1 - rhc$RHC) * w_treat * rhc$survival) / sum((1

```

```

## Warning in mean.default(rhc$survival[rhc$RHC == 1]): argument is not numeric or
## logical: returning NA

```

```

## Warning in Ops.factor((1 - rhc$RHC) * w_treat, rhc$survival): '*' not
## meaningful for factors

```

```

cat("ATT via IPW: ", round(tau_ATT_IPW, 3), "\n")

```

```

## ATT via IPW: NA

```

```

if (!requireNamespace("cobalt", quietly = TRUE)) install.packages("cobalt")
library(cobalt)

```

```

## cobalt (Version 4.6.2, Build Date: 2026-01-29)

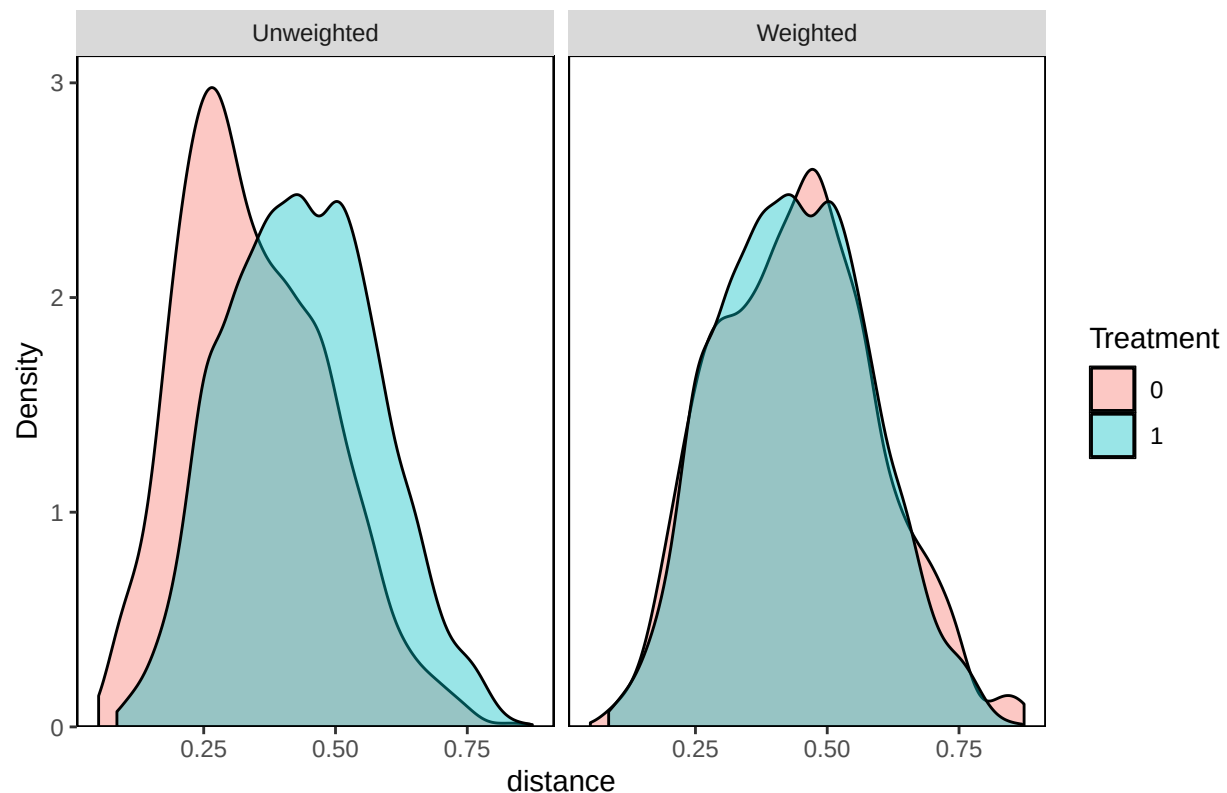
```

```

bal.plot(RHC ~ chfhx + aps1 + meanbp1 + paf11 + scoma1, data = rhc, weights = w_treat,
         distance = e_hat, method = "weighting", estimand = "ATT",
         var.name = "distance", which = "both", sample.names = c("Unweighted", "Weighted"))

```

Distributional Balance for "distance"



```
if (!requireNamespace("Matching", quietly=TRUE)) install.packages("Matching")
library(Matching)
```

```
## Loading required package: MASS
```

```
##
```

```
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
##      select
```

```
## ##
```

```
## ## Matching (Version 4.10-15, Build Date: 2024-10-14)
```

```
## ## See https://www.jsekhon.com for additional documentation.
```

```
## ## Please cite software as:
```

```
## ## Jasjeet S. Sekhon. 2011. "Multivariate and Propensity Score Matching
```

```
## ## Software with Automated Balance Optimization: The Matching package for R."
```

```
## ## Journal of Statistical Software, 42(7): 1-52.
```

```
## ##
```

```
m_att <- Match(Y = rhc$survival, Tr = rhc$RHC, X = e_hat,
  estimand = "ATT", M = 1, replace = TRUE)
tau_ATT_match <- m_att$est
```

```
cat("ATT via 1:1 PS matching: ", round(tau_ATT_match, 3), "\n")
```

```
## ATT via 1:1 PS matching: -0.027
```

```
tau_ATT_AIPW <- mean(
  rhc$RHC * (rhc$survival - mu0_hat) - (1 - rhc$RHC) *
  (e_hat / (1 - e_hat)) * (rhc$survival - mu0_hat)) / mean(rhc$RHC)
```

```
## Warning in Ops.factor(rhc$survival, mu0_hat): '-' not meaningful for factors
## Warning in Ops.factor(rhc$survival, mu0_hat): '-' not meaningful for factors
```

```
cat("ATT via AIPW (doubly robust): ", round(tau_ATT_AIPW, 3), "\n\n")
```

```
## ATT via AIPW (doubly robust): NA
```

```
one_rep <- function(covs_formula, df) {
  # standardization regression
  cat_vars <- c("aps1", "chfhx", "dnr1_Yes", "meanbp1")
  survival_form <- as.formula(paste("survival ~ ", covs_formula))

  fit_mu0 <- glm(survival_form, data = subset(df, RHC == 0), family=binomial)
  mu0_hat <- predict(fit_mu0, newdata = df, type="response")
  tau_ATT_reg <- mean(df$survival[df$RHC==1] - mu0_hat[df$RHC == 1])

  # Hajek IPW
  e_hat <- predict(glm(RHC ~ chfhx + aps1 + meanbp1 + dnr1_Yes, data = df, family =
    binomial), type = "response")

  w_treat <- df$RHC + (1 - df$RHC) * (e_hat / (1 - e_hat))
  tau_ATT_IPW <- mean(df$survival[df$RHC == 1]) - (sum((1 - df$RHC) * w_treat *
    df$survival) / sum((1-df$RHC)
    * w_treat))

  # Matching
  m_att <- Match(Y = df$survival, Tr = df$RHC, X = e_hat,
    estimand = "ATT", M = 1, replace = TRUE)
  tau_ATT_match <- m_att$est

  # AIPW (doubly robust)
  tau_ATT_AIPW <- mean(df$RHC * (df$survival - mu0_hat) - (1 - df$RHC) *
    (e_hat / (1 - e_hat)) * (df$survival - mu0_hat)) / mean(df$RHC)

  return(c(reg_via_std = round(tau_ATT_reg, 3),
    Hajek_IPW = round(tau_ATT_IPW, 3),
    Match = round(tau_ATT_match, 3),
    aipw = round(tau_ATT_AIPW, 3)))
}
```

```
set.seed(2026)
```

```

B <- 500
covs_formula <- paste(cat_vars, collapse = "+")

boot <- replicate(B, {
  idx <- sample(nrow(rhc), replace = TRUE)
  d <- rhc[idx, ]

  res <- suppressWarnings(one_rep(covs_formula, d))
  c(Reg = res[[1]],
    IPW = res[[2]],
    Match = res[[3]],
    AIPW = res[[4]])
})

# After populating: `boot` is a 4 x B matrix.
# 95% percentile CIs:
ci <- t(apply(boot, 1, quantile, probs = c(0.025, 0.975), na.rm=TRUE))

results_p1 <- data_frame(
  Estimator = c("Reg", "IPW", "Match", "AIPW"),
  Estimate = one_rep(covs_formula, rhc),
  CI_lower = ci[,1],
  CI_upper = ci[,2]
)

```

```

## Warning: 'data_frame()' was deprecated in tibble 1.1.0.
## i Please use 'tibble()' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

## Warning in Ops.factor(df$survival[df$RHC == 1], mu0_hat[df$RHC == 1]): '-' not
## meaningful for factors

## Warning in mean.default(df$survival[df$RHC == 1]): argument is not numeric or
## logical: returning NA

## Warning in Ops.factor((1 - df$RHC) * w_treat, df$survival): '*' not meaningful
## for factors

## Warning in Ops.factor(df$survival, mu0_hat): '-' not meaningful for factors
## Warning in Ops.factor(df$survival, mu0_hat): '-' not meaningful for factors

```

```
results_p1
```

```

## # A tibble: 4 x 4
##   Estimator Estimate CI_lower CI_upper
##   <chr>         <dbl>   <dbl>   <dbl>
## 1 Reg           NA       NA       NA
## 2 IPW           NA       NA       NA
## 3 Match        -0.027  -0.0560  0.006
## 4 AIPW          NA       NA       NA

```